COSC 341: Human Computer Interaction

Lab 4: Android Internal Storage, File Operations, and Firebase Database (20 marks)

The purpose of this lab is to get yourself familiar with Android internal storage, file operations, and Firebase database.

Configuration note:

- Please use Android Studio (with Java as the programming language—recommended platform for this course). The TAs will use this to load, build, and run your code. If you use any other toolkit or software development kit (e.g., Flutter), please write the corresponding name of the toolkit in the submission comment box in Canvas.
- Assignments will be marked using an Android Virtual Device (AVD) running a **Pixel 3a (API 33 or 35** is preferred).

Submission note:

Upload a zip file containing the following folders to Canvas: **Code** and **APK**.

- The **Code** folder should include the corresponding Android code.
- **The APK** folder should only contain one .apk file.

Also, in the submission comment box, please include **a video link showing** the app working on internal storage and Firebase.

You will implement an Android application that includes the following screens:

Main Screen [2 marks]

Your application should have a Main screen. This screen includes:

UI requirements [1 mark – deduct 0.5 for any missing widgets, a maximum deduction of 1 mark]

- A button showing the text "Read Data".
- A button showing the text "Write Data".
- A checkbox showing "Write on Database?"

Functional Requirement [1 mark – deduct 0.5 for any missing functionality, a maximum deduction of 1 mark]

When a user clicks on the

- "Read Data" button, the app opens the Read Data screen, as described later.
- "Write Data" button, the app opens the Write Data screen described in the next section.
- "Write on Database?" checkbox, the data will be logged on a Firebase Realtime Database.

Write Data Screen [8 marks]

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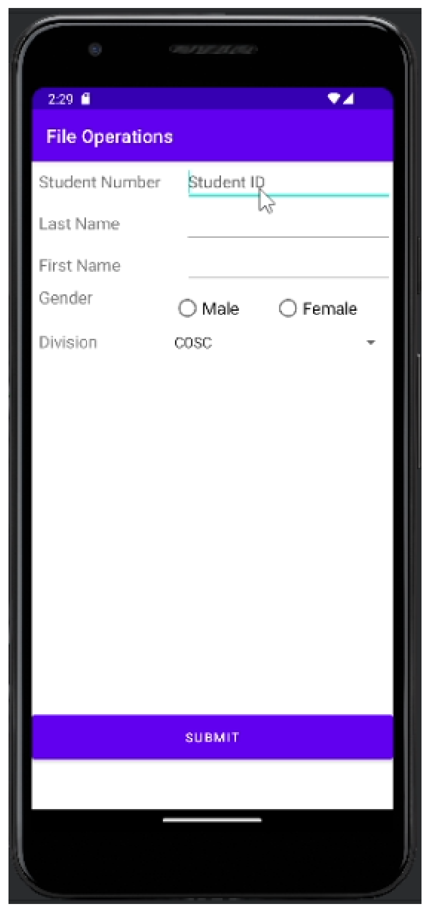
Your application should include a Write Data screen. Add the following UI components to allow a student to enter their information:

UI requirements [2 marks, 1.5 marks for widgets, 0.5 marks for validation with toast message – deduct 0.5 for any missing widgets, maximum deduction 1.5 marks]

- An edit text widget to enter an 8-digit student number
- An edit text widget to enter the last name
- Another edit text widget to enter the first name
- Radio buttons to enter gender information (At least two radio buttons)
- A dropdown list to select a division (at least three names, e.g., COSC, DATA, MATH should be on the list)
- A "Submit" button.
- [0.5 mark] If the student doesn't provide an 8-digit student number or doesn't enter the first and last name, show a Toast message asking to provide the correct information.

Main Screen



Write Data Screen

Functional Requirements [6 marks]

[3 marks] If the "Write on Database?" checkbox is not selected in the previous screen, the application should store data in a file. Therefore, once a user clicks on the submit button in this activity, the program will write the information/record (i.e., Student ID, Last Name, First Name, Gender, Division) into a file (e.g., data.txt), separated by commas (e.g., 12345678,hasan,khalad,Male,COSC). Additionally, each new record should be written in a new line (you can use "\n" to write a new record in a new line).

[3 marks] If the "Write on Database?" checkbox is selected in the previous screen, the application should store data in a Firebase Realtime database.

Once the information/record is written, the program will return to the Main Screen. This should be done using the finish() method, which will close the activity and return to the previous activity. While it is possible to return to the main screen by starting it as a new activity, this may result in a stack of activities.

Read Data Screen [7 marks]

The read Data screen will contain the following UI components:

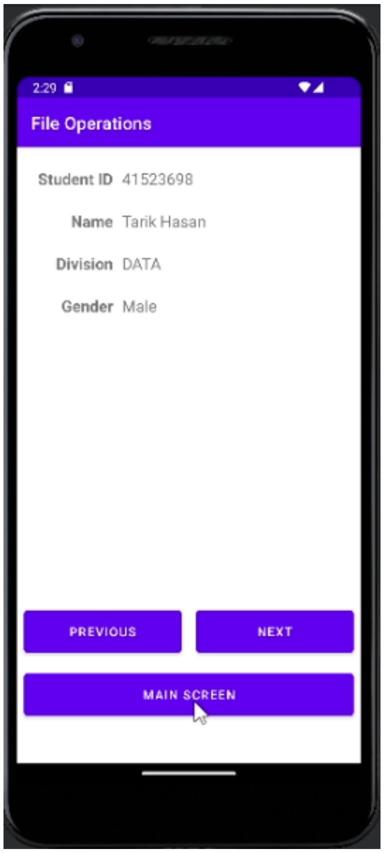
UI requirements [1 mark – deduct 0.5 for any missing widgets, a maximum deduction of 1 mark]

- A text view to show student number
- A text view to show the first name and the last name
- A text view to show gender information
- Another text view to show a division
- Two buttons, "Previous" and "Next"
- A "Main Screen" button.

Functional Requirements [6 marks]

[6 marks] Initially, the program will show the first record (e.g., Student Number: 12345678, Name: khalad hasan, Gender: Male, Division: COSC) from the file if "Write on Database?" is not selected or from the database if "Write on Database?" is selected. Note that if there is no record available (in the file or in the database), a Toast or some other message should be used to inform the user that they need to add some records. If the user presses the next or the previous button, it will show the next or previous record (if it's the last record, pressing the next button will show the first record. Similarly, if it's the first record, pressing the previous button would show the last record.

Additionally, clicking the Main Screen button would return to the main screen. This should be done by using a finish() method, not starting a new activity.



Video [3 Marks]

Create a video showing how the app works for the file system and database both for read and write operations. Upload the video to any video-sharing site (e.g., YouTube) and share the link.

Helpful Resources:

Here is a sample video showing the app working: **https://youtu.be/CNPpMnRL3VE**